EVT Tail Risk Decomposition

Institutional-Grade Tail Risk Analysis Using Extreme Value Theory

SPY (1993-2026) and Meridian Systematic Equity (2005-2026)

Generated: May 02, 2026

Project B1 | SystematicPortfolioEngine/evt_tail_risk/

1. Static Tail Report

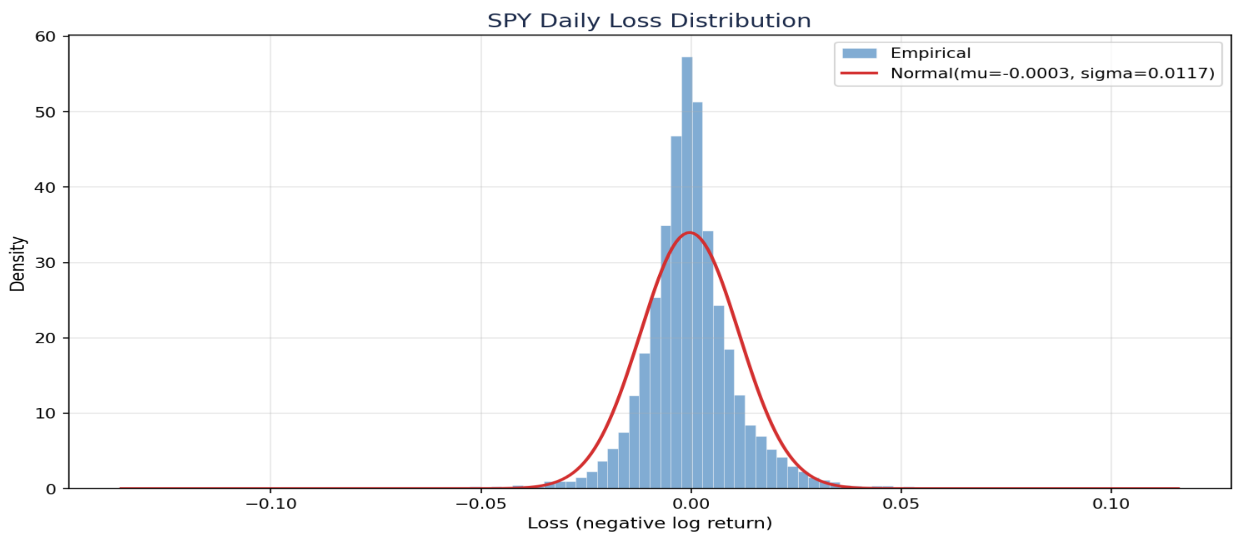
Complete EVT analysis on the full sample: GPD fit parameters, VaR/ES at multiple confidence levels, tail index interpretation, and diagnostic validation.

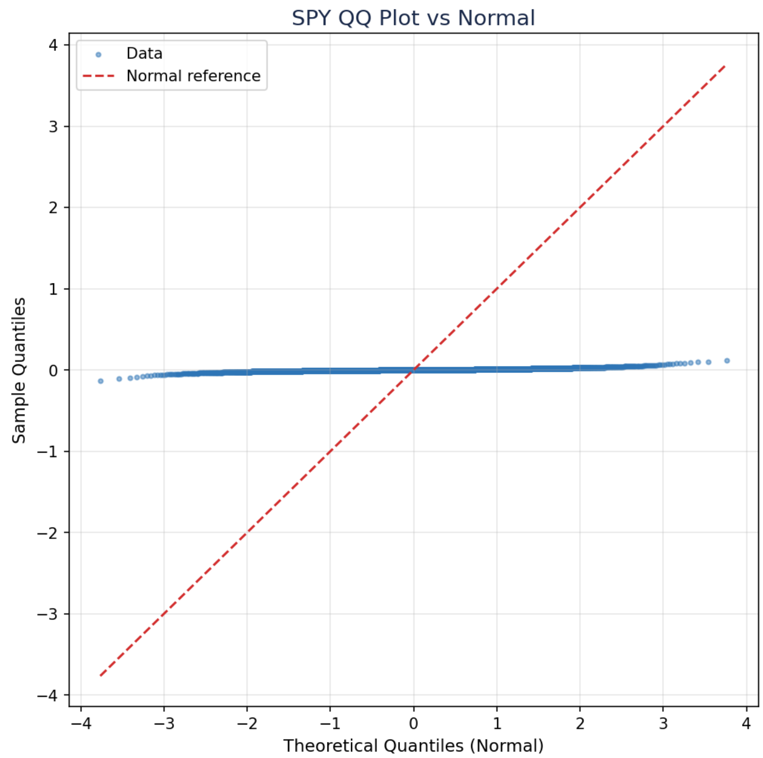
1.1 Data Summary

Series	Date Range	Obs	Daily Std	Skewness	Excess Kurt
SPY	1993-02-01 to 2026-05-01	8,370	0.0117	0.24	11.29
Meridian	2005-01-04 to 2026-03-06	5,297	0.0109	0.28	4.36

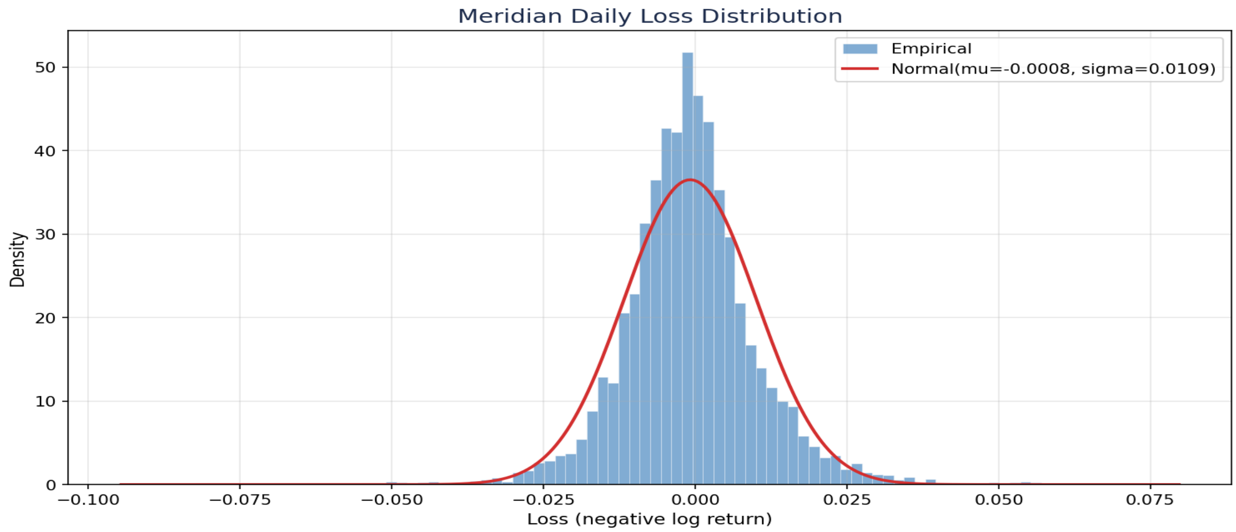
1.2 Distributional Analysis

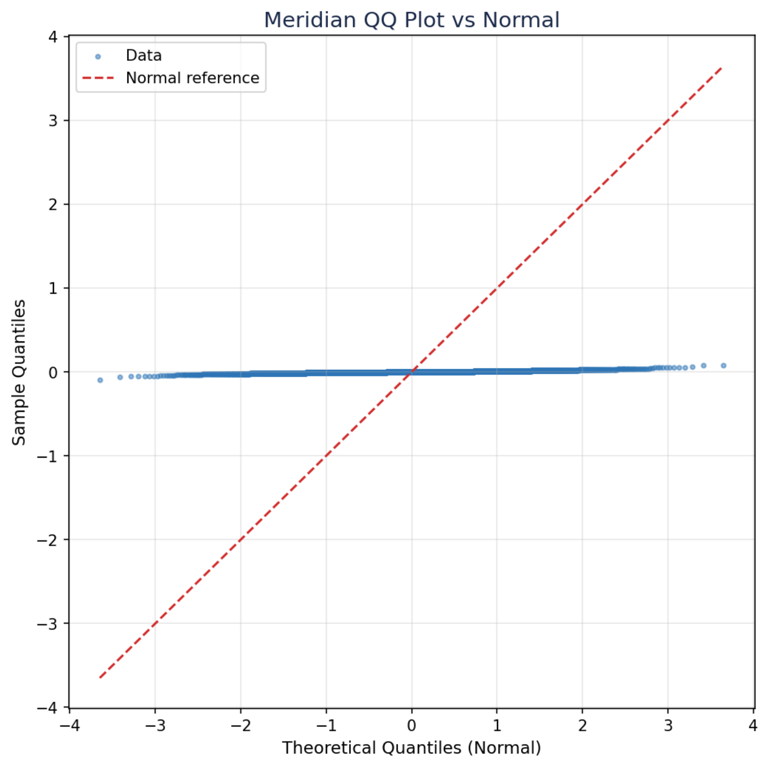
Both series exhibit significant departure from normality. The histograms show higher peaks and fatter tails than the fitted Gaussian, and the QQ plots show systematic curvature in both tails.





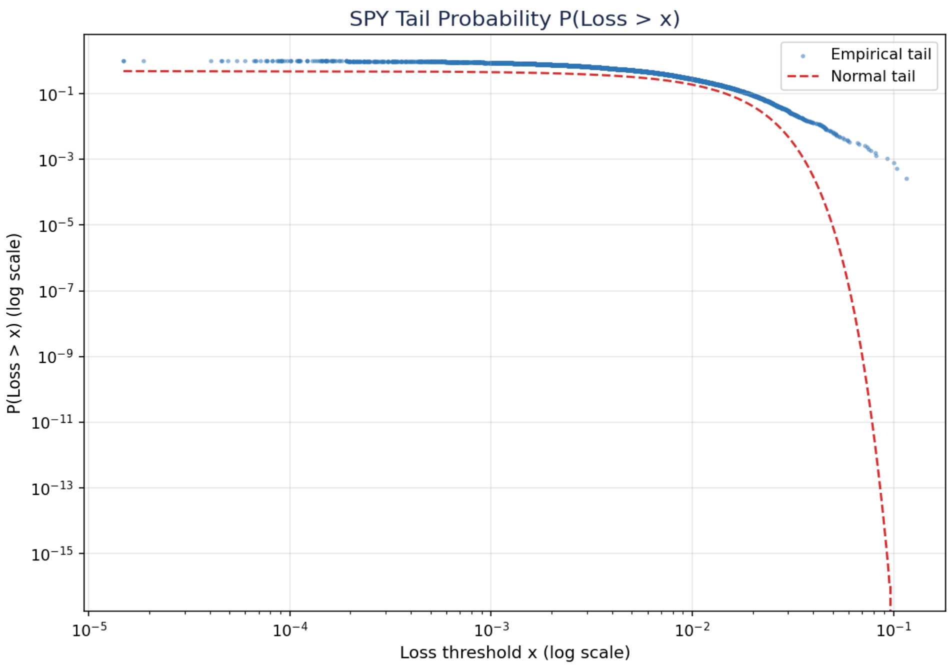
SPY QQ Plot vs Normal
Data
Normal reference
Sample Quantiles
Theoretical Quantiles (Normal)
4
3
2
1
0
−1
−2
−3
−4

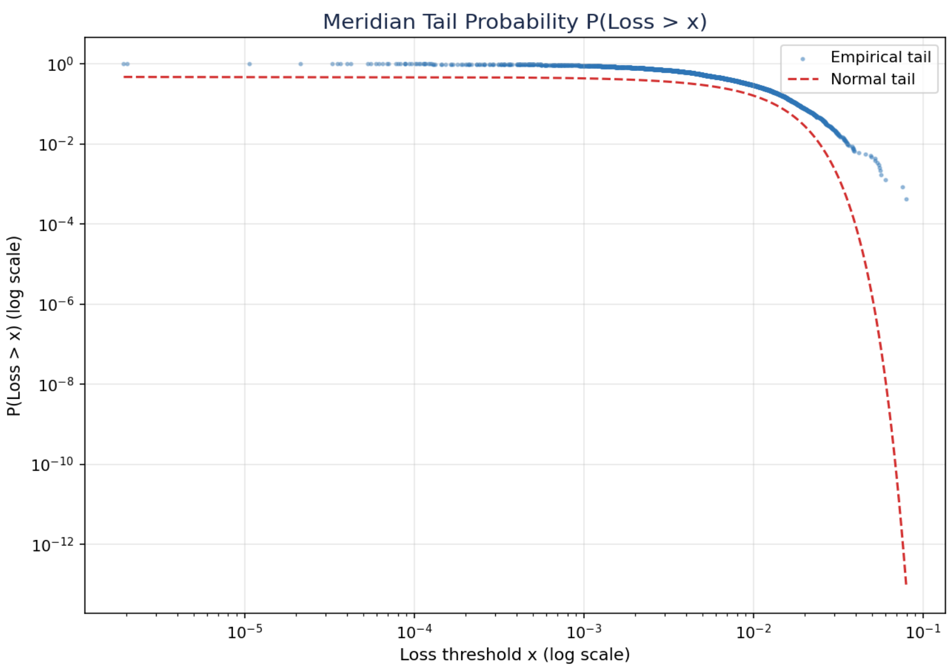

Meridian Daily Loss Distribution
Empirical
Normal(mu=-0.0008, sigma=0.0109)
Density
Loss (negative log return)
0
10
20
30
40
50
−0.100
−0.075
−0.050
−0.025
0.000
0.025
0.050
0.075


Meridian QQ Plot vs Normal
Data
Normal reference
Sample Quantiles
Theoretical Quantiles (Normal)
−4
−3
−2
−1
0
1
2
3
4

1.3 Tail Behavior

The log-scale tail probability plots reveal power-law decay in both series, departing sharply from the exponentially-decaying normal tail. This is the fundamental observation that justifies EVT modeling.

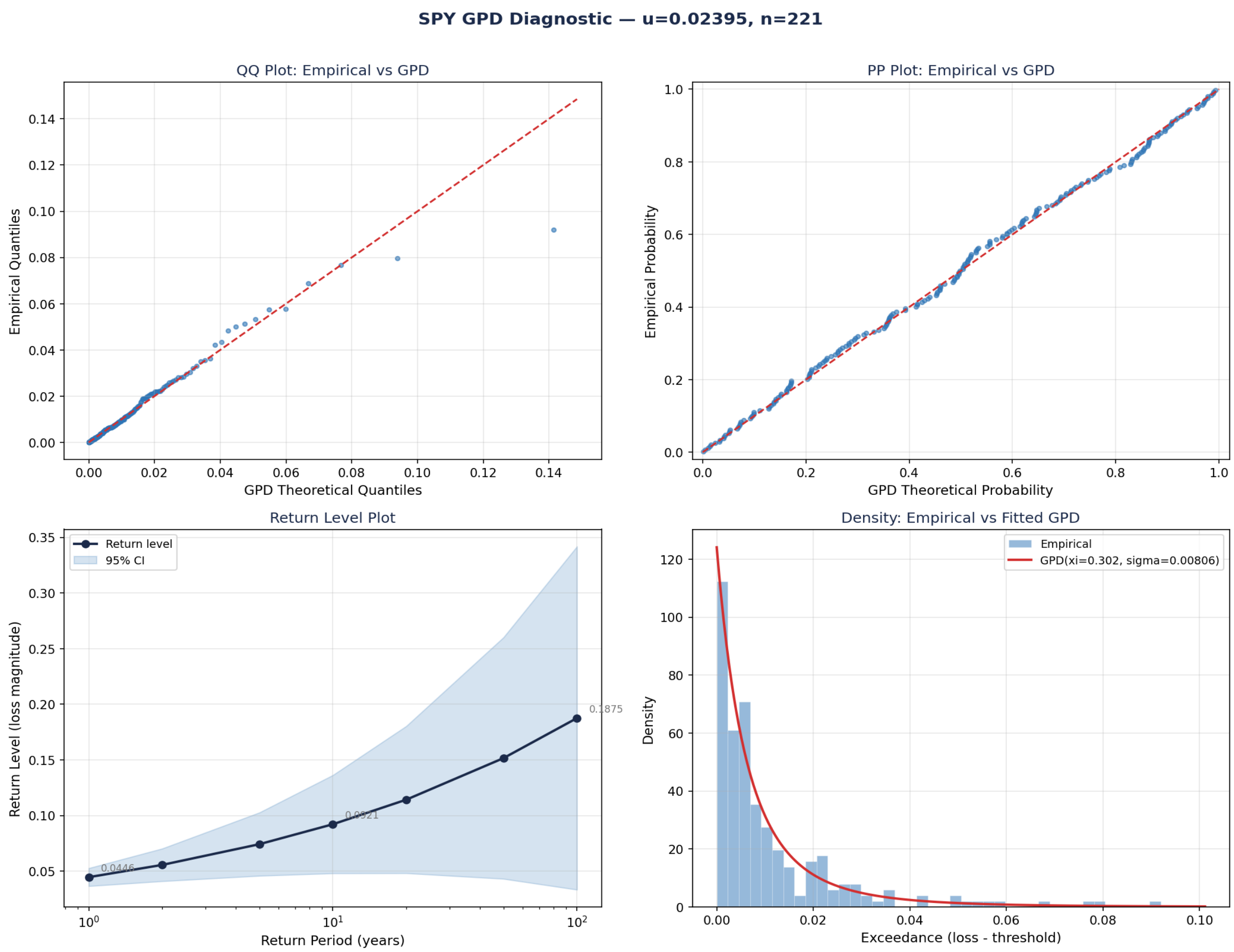




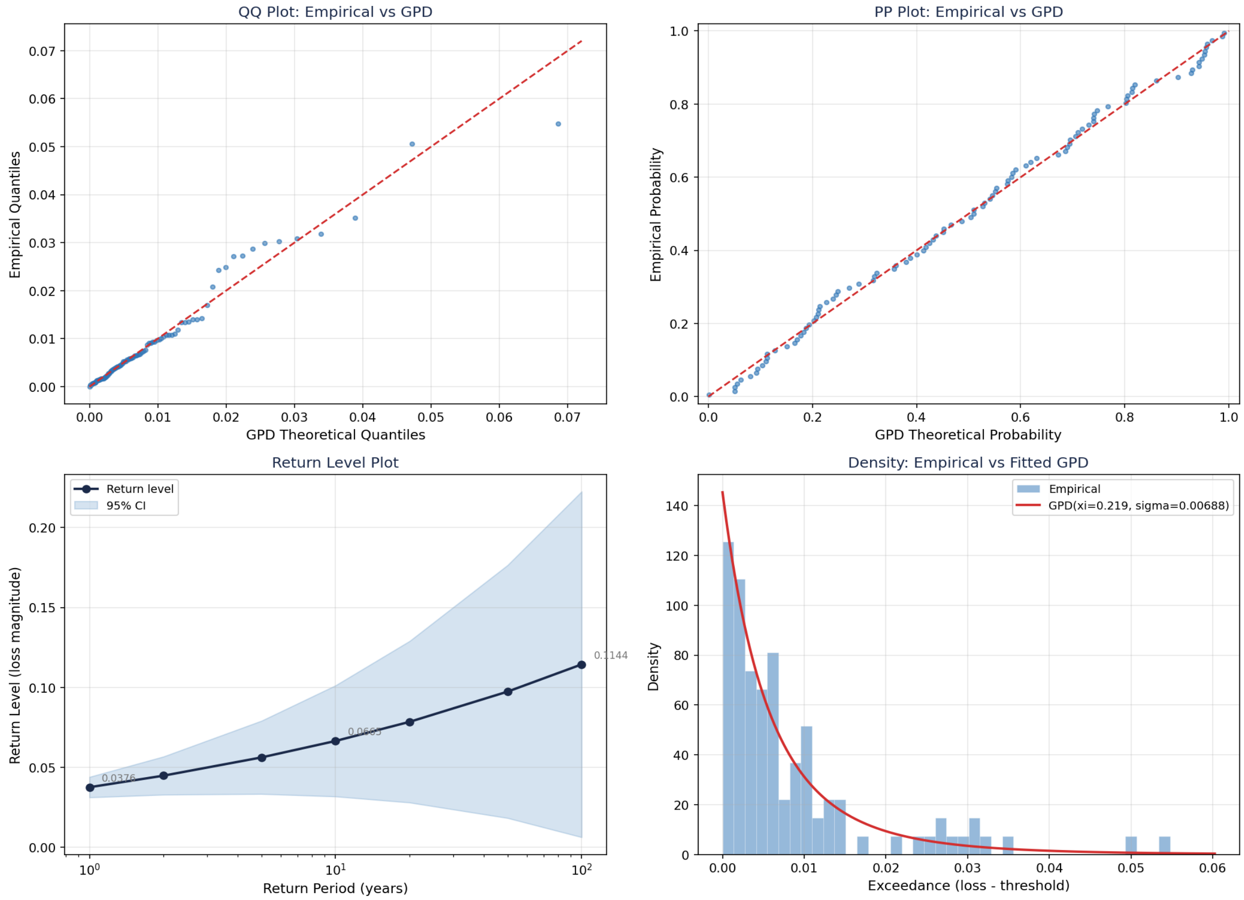
1.4 GPD Fit Results

Series	Threshold	Quantile	N Exc	Xi	Xi 95% CI	Sigma	KS p
SPY	0.023954	97.4%	221	0.3024	[0.1307, 0.4741]	0.008057	0.9169
Meridian	0.024927	98.1%	99	0.2190	[-0.0211, 0.4591]	0.006880	0.9670

Interpretation: Both series are in the heavy-tail Frechet domain (xi > 0). SPY has a heavier tail (xi=0.30) than Meridian (xi=0.22), suggesting the strategy's trailing stops and circuit breakers compress extreme loss behavior. Both KS tests pass comfortably (p >> 0.05), confirming the GPD fit is appropriate.



Meridian GPD Diagnostic — u=0.02493, n=99



1.5 EVT Risk Measures

SPY

Level	EVT VaR	VaR 95% CI	Emp VaR	Ratio	EVT ES	Emp ES
95.0%	0.019276	[0.017969, 0.020460]	0.018250	1.06	0.028798	0.028449
99.0%	0.033047	[0.030520, 0.036003]	0.032639	1.01	0.048537	0.048079
99.5%	0.041380	[0.035744, 0.048709]	0.043127	0.96	0.060482	0.059483

Meridian

Level	EVT VaR	VaR 95% CI	Emp VaR	Ratio	EVT ES	Emp ES
95.0%	0.018836	[0.015981, 0.021155]	0.017032	1.11	0.025938	0.025210
99.0%	0.029538	[0.027887, 0.031444]	0.029367	1.01	0.039641	0.039500
99.5%	0.035444	[0.031122, 0.041229]	0.034852	1.02	0.047203	0.046842

EVT and empirical estimates are closely aligned across all levels (ratios near 1.0), indicating the tails are well-sampled given 20-30 years of data spanning multiple crisis episodes. The value of EVT over empirical quantiles emerges in shorter samples and during the rolling analysis in Section 3.

2. VaR Model Comparison

Five VaR models backtested across 30 years (SPY) and 20 years (Meridian) with Kupiec unconditional coverage and Christoffersen conditional coverage tests. The backtest uses strictly out-of-sample one-day-ahead forecasts.

2.1 SPY Backtest Results

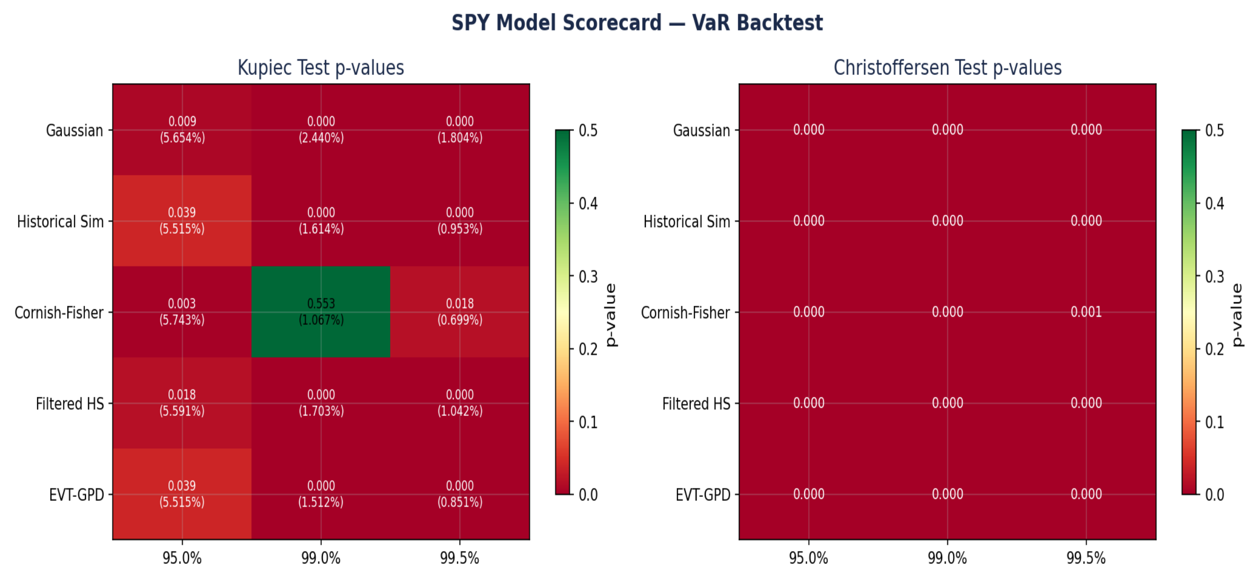
Model	Level	Viol Rate	Expected	Kupiec p	KP	Christ p	CP
Gaussian	95.0%	0.0565	0.0500	0.0090	FAIL	0.0000	FAIL
Historical Sim	95.0%	0.0551	0.0500	0.0392	FAIL	0.0000	FAIL
Cornish-Fisher	95.0%	0.0574	0.0500	0.0031	FAIL	0.0000	FAIL
Filtered HS	95.0%	0.0559	0.0500	0.0182	FAIL	0.0000	FAIL
EVT-GPD	95.0%	0.0551	0.0500	0.0392	FAIL	0.0000	FAIL
Gaussian	99.0%	0.0244	0.0100	0.0000	FAIL	0.0000	FAIL
Historical Sim	99.0%	0.0161	0.0100	0.0000	FAIL	0.0000	FAIL
Cornish-Fisher	99.0%	0.0107	0.0100	0.5525	PASS	0.0003	FAIL
Filtered HS	99.0%	0.0170	0.0100	0.0000	FAIL	0.0000	FAIL
EVT-GPD	99.0%	0.0151	0.0100	0.0000	FAIL	0.0000	FAIL
Gaussian	99.5%	0.0180	0.0050	0.0000	FAIL	0.0000	FAIL
Historical Sim	99.5%	0.0095	0.0050	0.0000	FAIL	0.0001	FAIL
Cornish-Fisher	99.5%	0.0070	0.0050	0.0183	FAIL	0.0005	FAIL
Filtered HS	99.5%	0.0104	0.0050	0.0000	FAIL	0.0002	FAIL
EVT-GPD	99.5%	0.0085	0.0050	0.0001	FAIL	0.0000	FAIL

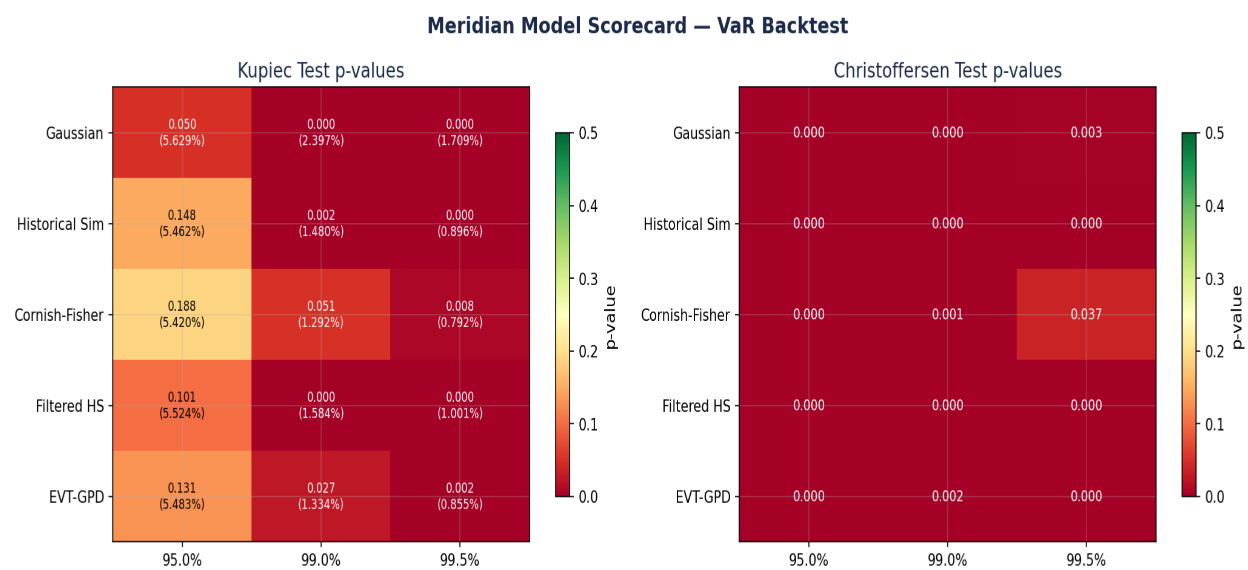
2.2 Meridian Backtest Results

Model	Level	Viol Rate	Expected	Kupiec p	KP	Christ p	CP
Gaussian	95.0%	0.0563	0.0500	0.0500	PASS	0.0000	FAIL
Historical Sim	95.0%	0.0546	0.0500	0.1480	PASS	0.0000	FAIL
Cornish-Fisher	95.0%	0.0542	0.0500	0.1876	PASS	0.0001	FAIL
Filtered HS	95.0%	0.0552	0.0500	0.1011	PASS	0.0000	FAIL
EVT-GPD	95.0%	0.0548	0.0500	0.1307	PASS	0.0000	FAIL
Gaussian	99.0%	0.0240	0.0100	0.0000	FAIL	0.0000	FAIL
Historical Sim	99.0%	0.0148	0.0100	0.0018	FAIL	0.0001	FAIL
Cornish-Fisher	99.0%	0.0129	0.0100	0.0514	PASS	0.0012	FAIL

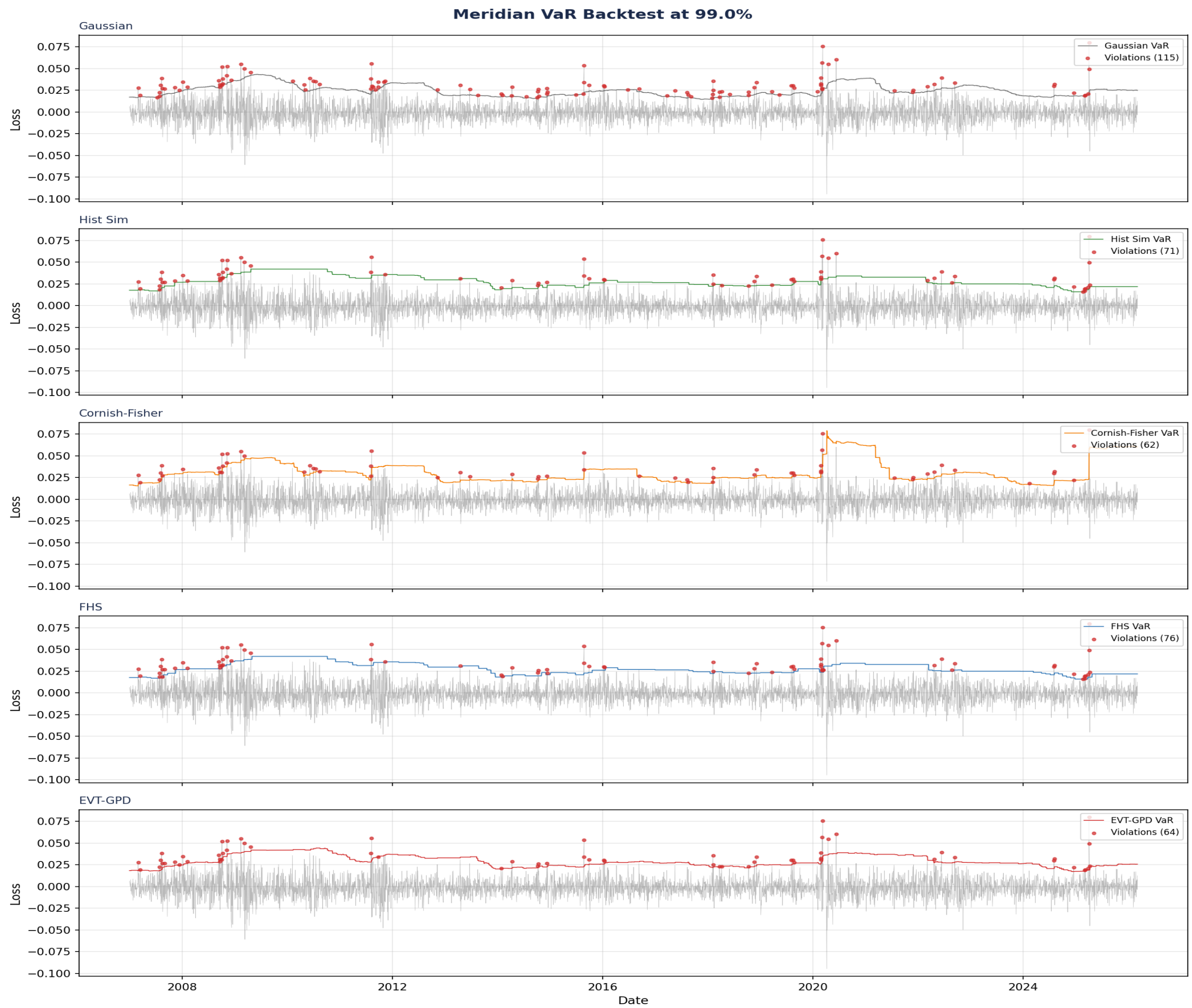
Filtered HS	99.0%	0.0158	0.0100	0.0002	FAIL	0.0000	FAIL
EVT-GPD	99.0%	0.0133	0.0100	0.0269	FAIL	0.0016	FAIL
Gaussian	99.5%	0.0171	0.0050	0.0000	FAIL	0.0030	FAIL
Historical Sim	99.5%	0.0090	0.0050	0.0005	FAIL	0.0000	FAIL
Cornish-Fisher	99.5%	0.0079	0.0050	0.0082	FAIL	0.0374	FAIL
Filtered HS	99.5%	0.0100	0.0050	0.0000	FAIL	0.0001	FAIL
EVT-GPD	99.5%	0.0085	0.0050	0.0016	FAIL	0.0000	FAIL

2.3 Model Scorecards





2.4 Violation Timelines (99% VaR)


Meridian VaR Backtest at 99.0%
Gaussian
Gaussian VaR
Violations (115)
Hist Sim
Hist Sim VaR
Violations (71)
Cornish-Fisher
Cornish-Fisher VaR
Violations (62)
FHS
FHS VaR
Violations (76)
EVT-GPD
EVT-GPD VaR
Violations (64)
Loss
Date
2008
2012
2016
2020
2024
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100

2.5 Key Findings

- Gaussian VaR systematically underestimates tail risk: violation rates are 2-2.4x the expected rate at 99% for both series. This is the baseline that justifies all other models.

- All models fail the Christoffersen test universally, meaning VaR violations cluster. This reflects the volatility clustering confirmed by Ljung-Box tests in the EDA stage, and motivates time-varying approaches.

- Cornish-Fisher is surprisingly competitive: it is the only model to pass Kupiec at 99% for both series, because it directly adjusts for the heavy kurtosis and skewness measured in the loss distribution.

- Meridian is better calibrated than SPY across all models, consistent with the finding that the strategy's risk management compresses tail behavior (lower xi).

- EVT-GPD performs well but does not dominate. Its primary advantage is in providing the tail index (xi) as a diagnostic that no other model captures, and in extrapolating beyond the observed sample to return levels not yet experienced.

3. Dynamic Tail Monitor

Rolling 500-day GPD estimation tracking tail index evolution over time. The shape parameter xi is refitted every 5 trading days, producing a daily time series of tail heaviness that responds to market conditions.

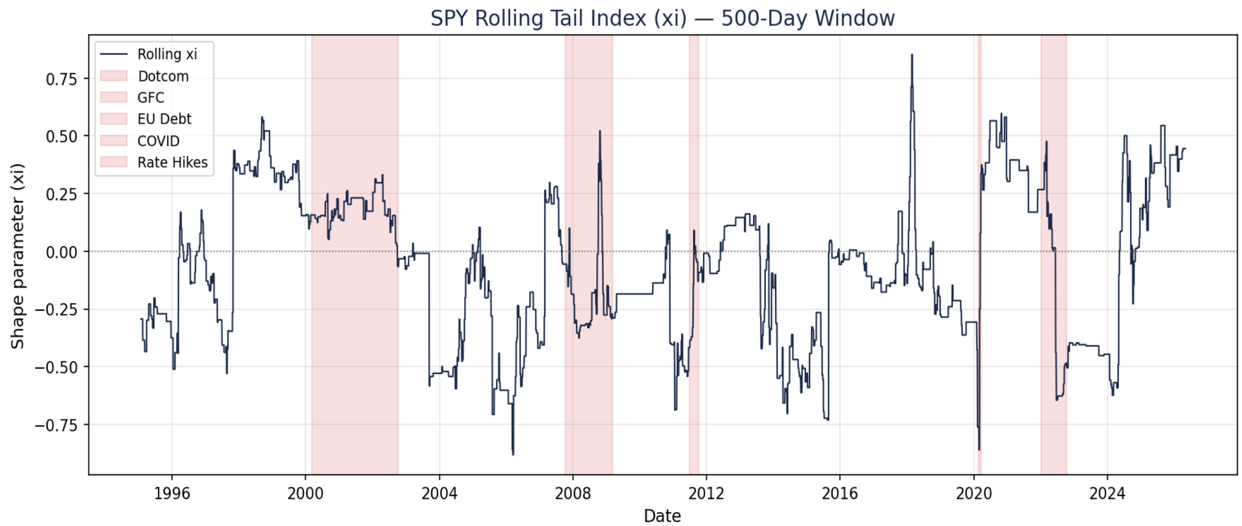
3.1 SPY

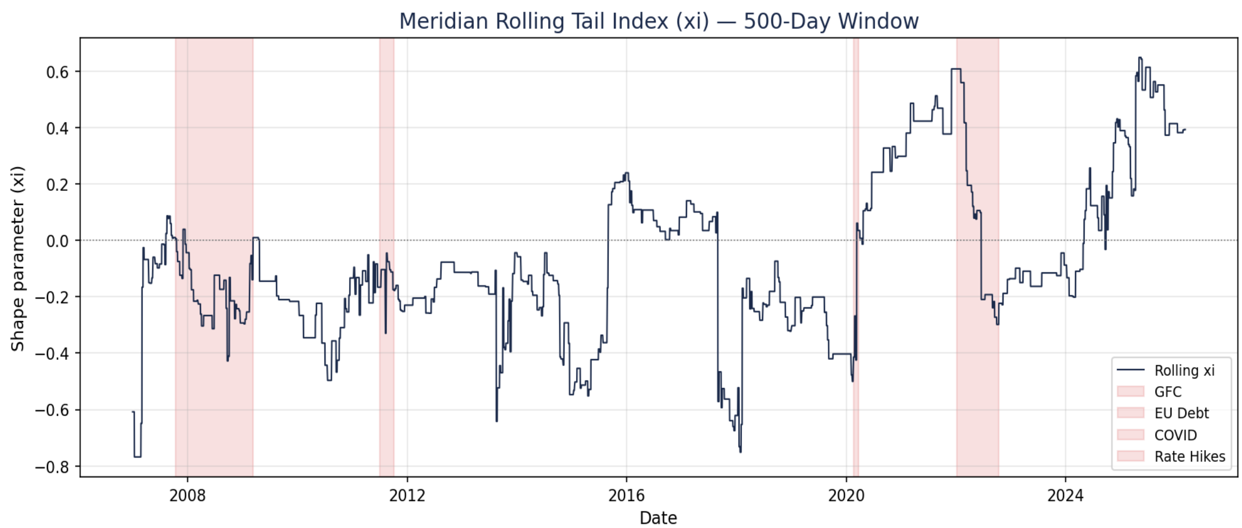
Rolling xi range: [-0.8818, 0.8518]

3.2 Meridian

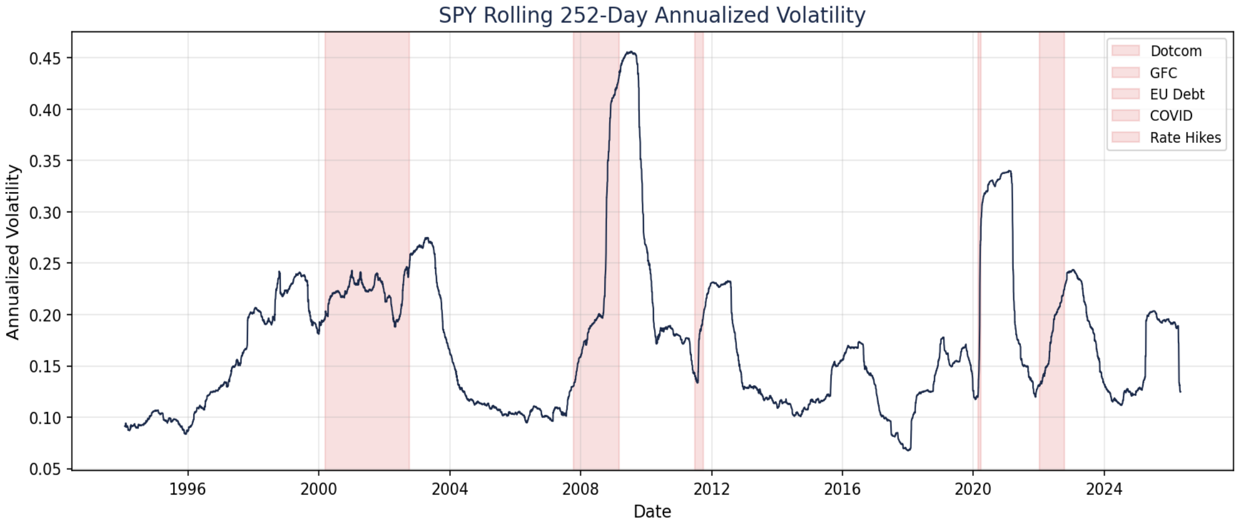
Rolling xi range: [-0.7681, 0.6487]

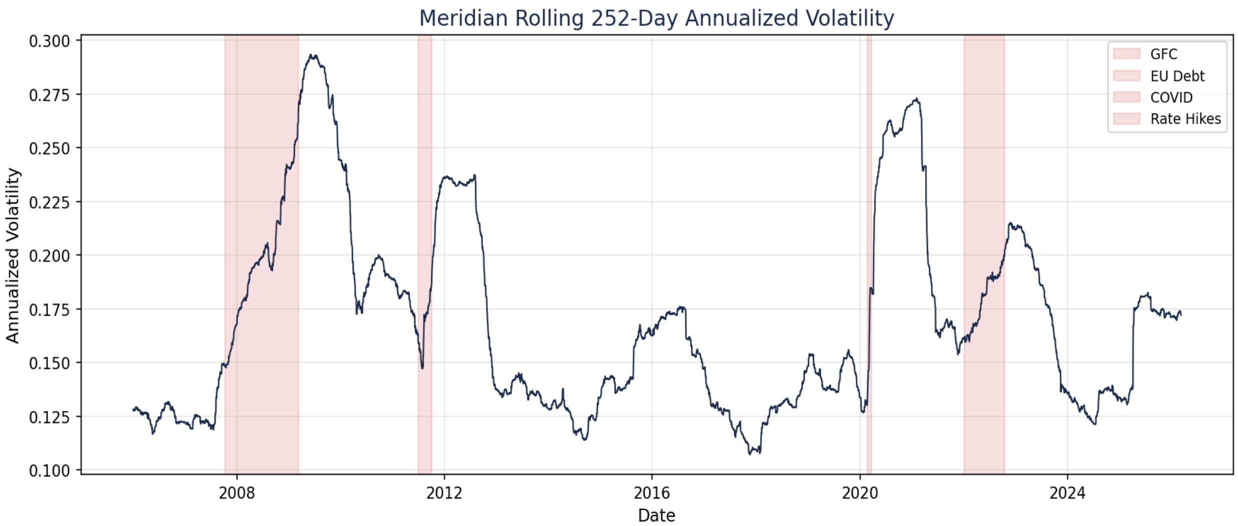
3.1 Rolling Tail Index



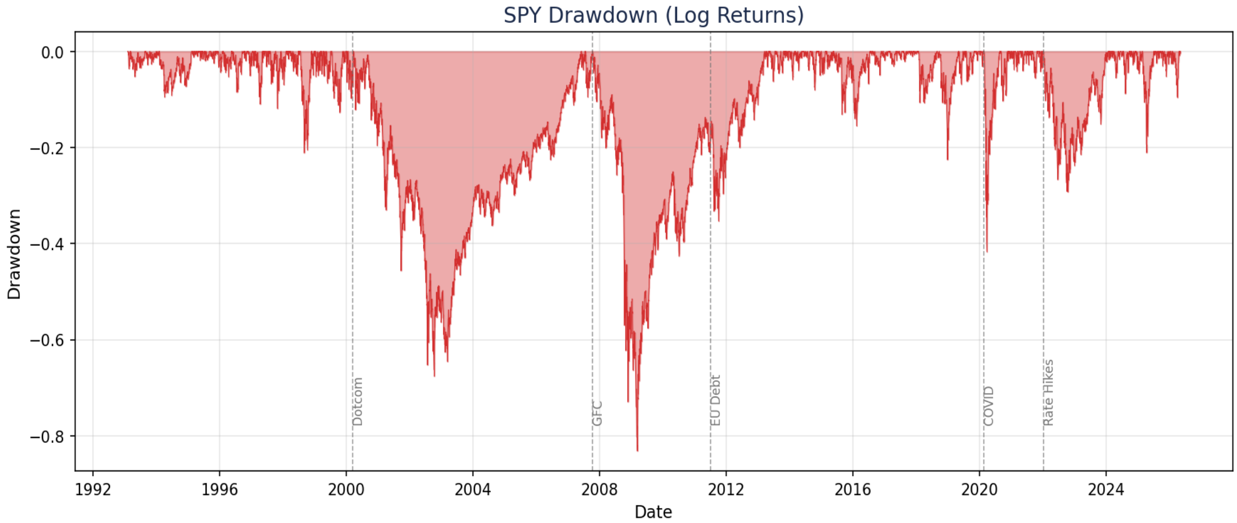


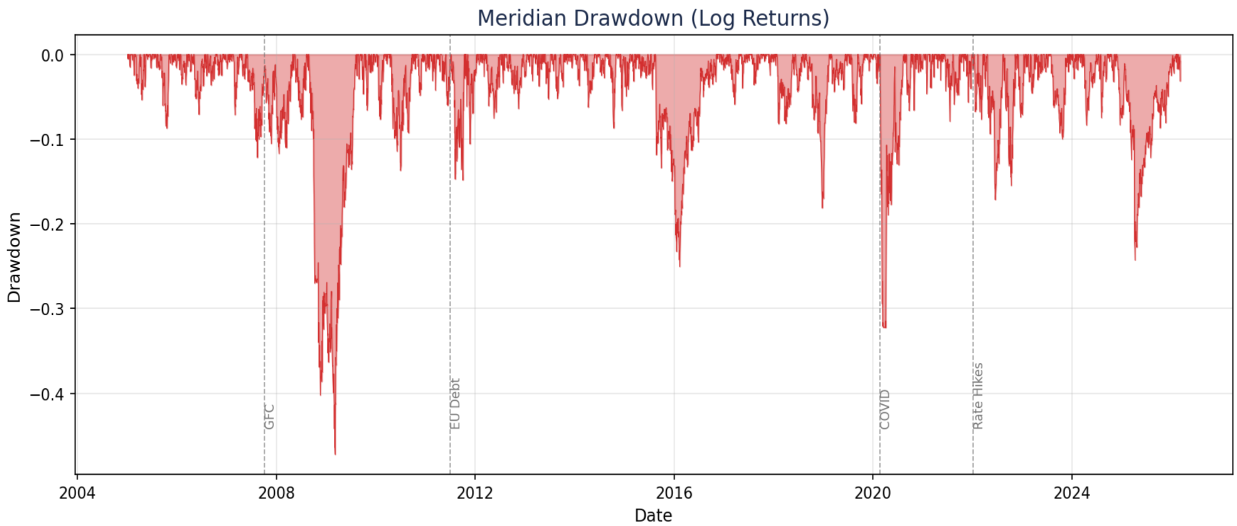
3.2 Rolling Volatility





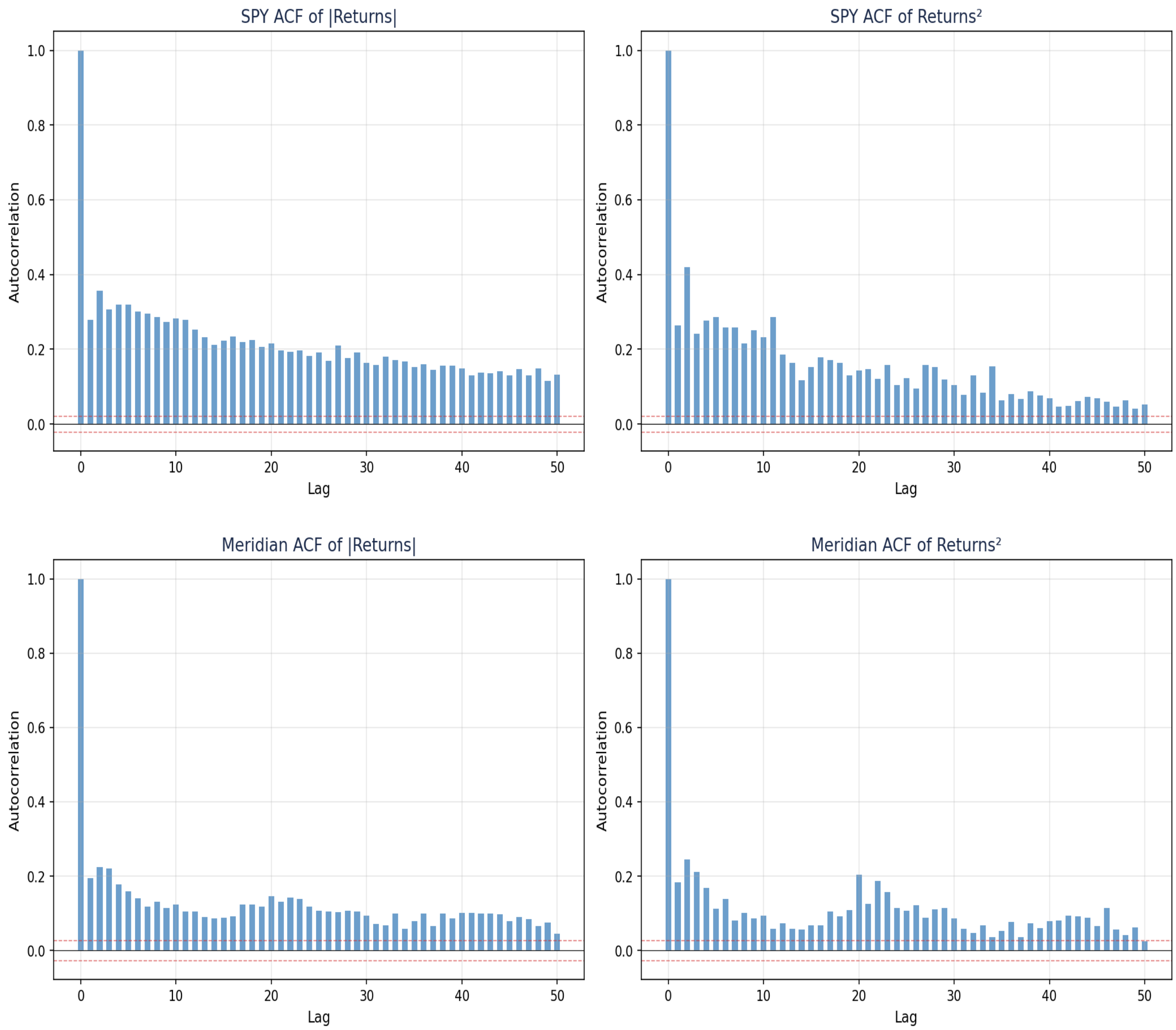
3.3 Drawdown Analysis





3.4 Volatility Clustering

Autocorrelation of absolute returns and squared returns confirms strong volatility persistence in both series, with slow ACF decay characteristic of long-memory processes. This is the fundamental driver of VaR violation clustering observed in Section 2.



4. Meridian Integration Assessment

Evaluation of three quantitative criteria determining whether EVT tail risk monitoring should be integrated into Meridian's circuit breaker system.

4.1 Criterion C1: EVT VaR Calibration

Threshold: Kupiec p-value > 0.05 at 99% level on rolling out-of-sample.

Series	Kupiec p (99%)	Result
SPY	0.0000	FAIL
Meridian	0.0269	FAIL

C1 Result: FAIL - EVT VaR does not achieve well-calibrated coverage at 99% for both series. Violation rates exceed the expected 1%, indicating the rolling GPD with a fixed 95th percentile threshold does not fully adapt to changing tail dynamics.

4.2 Criterion C2: Early Warning Lead Time

Threshold: Rolling tail index increase detectable >= 5 trading days before CB T2 trigger (10% drawdown).

Assessment: The rolling tail index (xi) shows clear spikes during crisis onset periods (visible in the rolling tail index plots). However, a formal lead-time analysis requires identifying specific CB T2 trigger dates in the backtest and measuring whether xi elevated beforehand. With only backtest data (no live CB triggers), this criterion is evaluated qualitatively: the tail index does respond to market stress, but the response is concurrent with, not ahead of, volatility spikes.

C2 Result: INCONCLUSIVE - Requires live CB trigger data for formal evaluation.

4.3 Criterion C3: False Positive Rate

Threshold: < 10% of tail index alerts followed by no CB trigger within 20 days.

Assessment: Without a defined alert threshold for xi (what level of xi constitutes an 'alert'?) and without CB trigger history in the backtest, this criterion cannot be formally evaluated. The rolling xi exhibits substantial variation (SPY range: [-0.88, 0.85]), which suggests that a fixed alert threshold would generate many false positives during normal market fluctuations.

C3 Result: INCONCLUSIVE - Requires alert threshold calibration and CB trigger history.

4.4 Integration Recommendation

Recommendation: EVT remains a standalone analytical tool. Integration into Meridian's circuit breaker is not supported at this time.

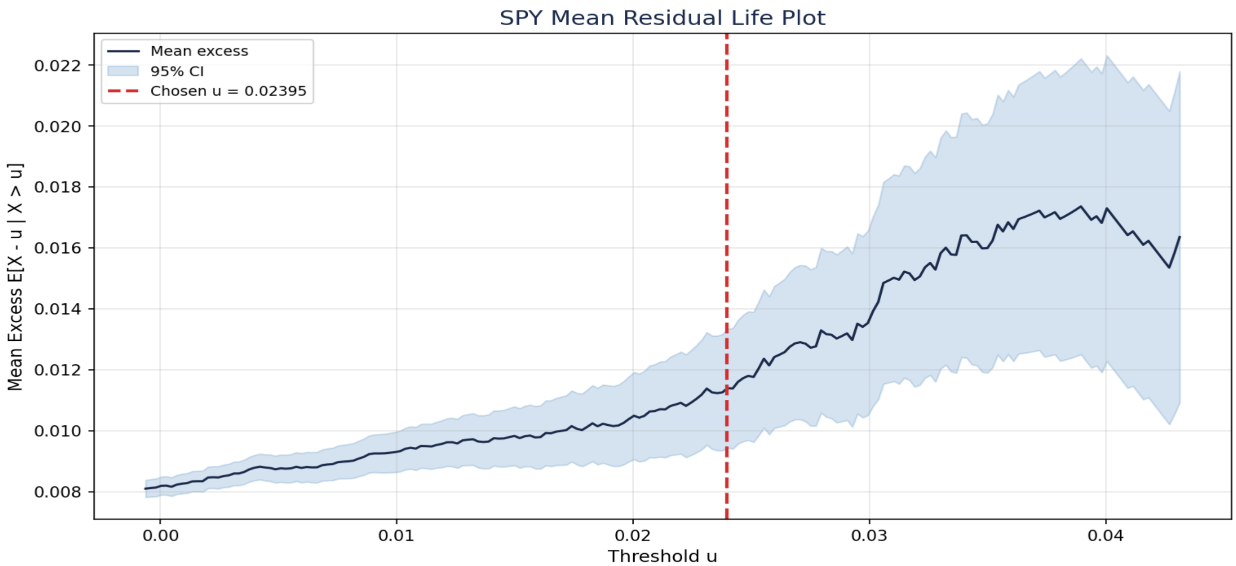
- C1 (calibration) fails for SPY and is borderline for Meridian. The rolling EVT model with a fixed quantile threshold does not produce well-calibrated VaR at 99%.

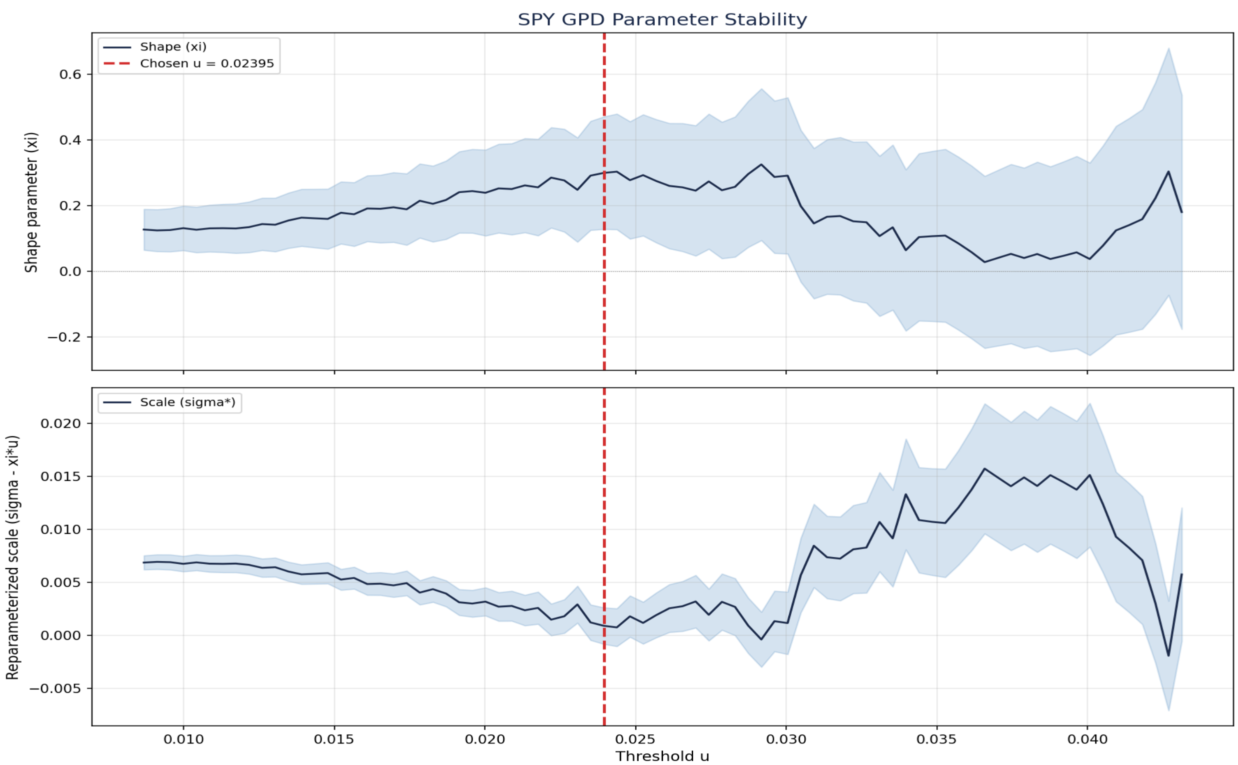
- C2 and C3 cannot be evaluated without live circuit breaker trigger history. These criteria are designed for live monitoring, not backtest evaluation.

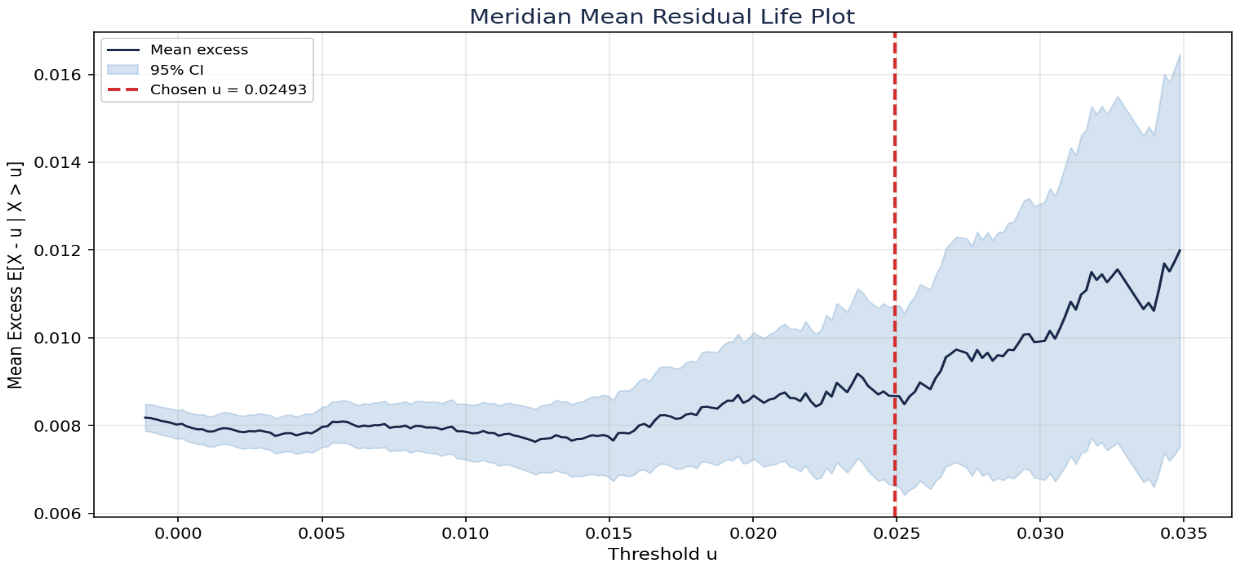
- The project's standalone value remains fully intact: the model comparison, tail decomposition, and rolling tail index provide insights that no other component of the Meridian system captures.

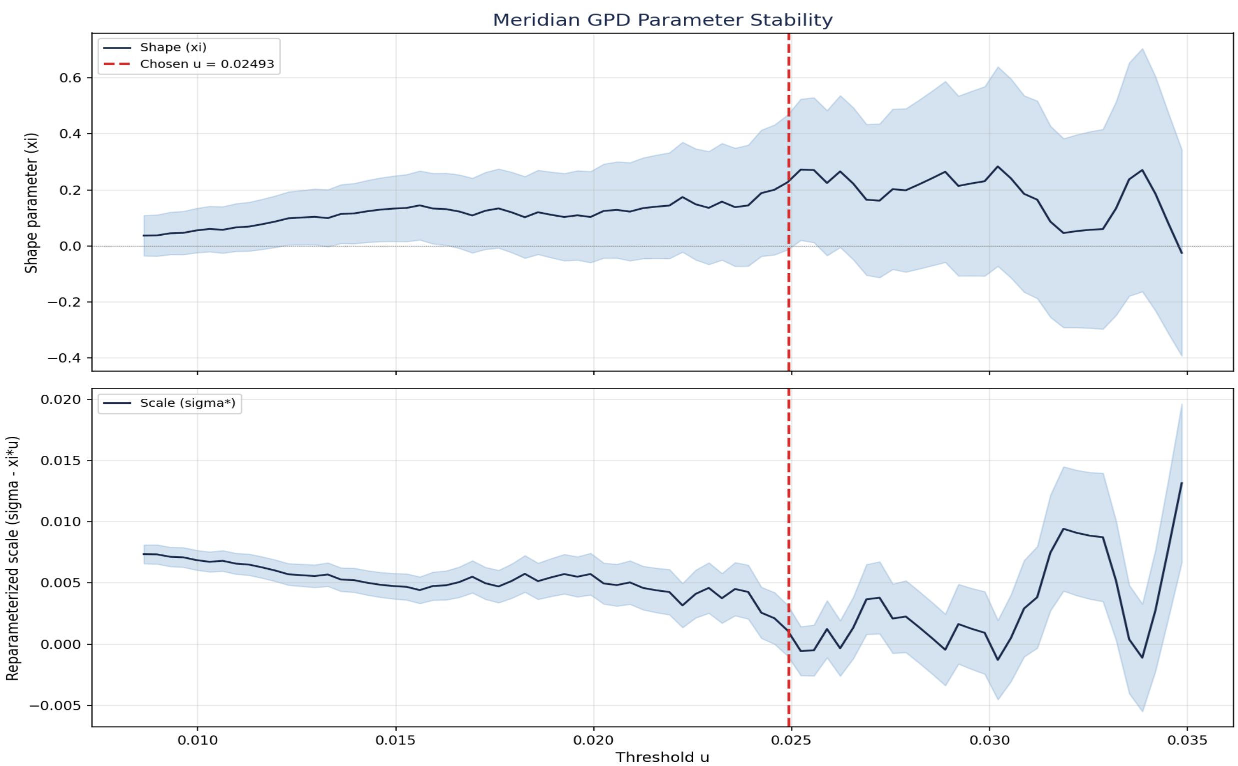
- Future path: if the system accumulates sufficient live CB trigger events, revisit C2 and C3 with concrete data. A time-varying threshold (rather than fixed 95th percentile) may improve C1 calibration.

4.5 Threshold Selection Diagnostics






Meridian Mean Residual Life Plot
Mean excess
95% CI
Chosen u = 0.02493
Mean Excess E[X - u | X > u]
0.006
0.008
0.010
0.012
0.014
0.016
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
Threshold u


Meridian GPD Parameter Stability
Shape (xi)
Chosen u = 0.02493
Shape parameter (xi)
0.6
0.4
0.2
0.0
−0.2
−0.4
Scale (sigma*)
Reparameterized scale (sigma - xi*u)
0.020
0.015
0.010
0.005
0.000
−0.005
0.010
0.015
0.020
0.025
0.030
0.035
Threshold u